Creating a Golden Image

Building, capturing, verifying and publishing a Windows image for the ALS audit stations.

Image server	**192.168.0.20** (user `stephen01`)
Staging folder	`/srv/als-incoming` writable - captures land here
Image library	`/srv/als-images` read-only to stations
Seen by stations as	`/home/partimag` on the audit station

The whole job in one line

Build a machine → sysprep it → capture it with Clonezilla to the staging folder → verify the capture → move it into the library → name it on a station. Roughly 2 to 3 hours, most of which is unattended.

> **Why the library is read-only**
> Stations mount it read-only on purpose. A machine being wiped or imaged can never corrupt or encrypt your golden images, even if that machine is compromised. This is why captures land in a separate writable folder and are moved in afterwards.

Before you start

- A machine to build the image on. Its disk should be **no larger than** the smallest machine you will restore onto. An image captured from a 256 GB disk will not restore onto a 128 GB one.
- A wired network cable to the image server. Over Wi-Fi a 15 GB capture takes 20-30 minutes instead of about 2 and a half.
- A Clonezilla boot USB (not the audit stick - a plain Clonezilla Live stick).
- Free space on the server: at least 1.5× the size of the image you are about to make.
- Your Windows licence / activation approach settled before you capture, not after.

```
ssh stephen01@192.168.0.20
df -h /srv
```

Look at the **Avail** column for the filesystem holding `/srv`. A Windows 11 image typically compresses to 12-18 GB.

Step 1 · Build the reference machine

Install and configure one machine exactly as you want every customer to receive it. Everything you do here is copied to every future machine, and anything you forget has to be done by hand on all of them.

Do install

- Windows, fully updated.
- Chipset, graphics, network and audio drivers.
- Your standard software set.
- Any default settings, power plan, or branding you want shipped.

Do not install

- Anything tied to one machine's hardware - a driver for a graphics card only that model has.
- Anything licensed per-machine that cannot be re-activated.
- Personal accounts. Sign out of everything.

> **Leave the disk smaller than you think you need**
> Clonezilla restores a disk image onto a disk of the same size or larger, never smaller. Build on the smallest capacity you stock. The station refuses a target that is too small, but only after you have started.

Step 2 · Sysprep — do not skip this

Sysprep strips the machine-specific identity: the security ID, the computer name, the hardware-tied activation state. Without it every machine you ship carries the same identity, which breaks Windows Update on some units, breaks domain joins, and is not something you want to discover after shipping fifty machines.

On the reference machine, in an Administrator Command Prompt:

```
C:\Windows\System32\Sysprep\sysprep.exe /generalize /oobe /shutdown
```

The machine shuts itself down when it finishes. **Do not switch it back on** - booting Windows again undoes the generalize step and you would have to run it a second time. Go straight to the capture.

> **If sysprep refuses to run**
> Usually a Microsoft Store app blocking generalize. The log at `C:\Windows\System32\Sysprep\Panther\setuperr.log` names the package. Remove it for all users, then run sysprep again. Do not work around it by skipping /generalize.

Step 3 · Capture the image to the server

Boot the reference machine from the **Clonezilla** USB, with the network cable plugged in.

1. Language and keyboard - accept the defaults.
2. **Start_Clonezilla**
3. **device-image**
4. **nfs_server**
5. Network - **dhcp**
6. NFS version - **nfs** (v3)
7. Server IP - **`192.168.0.20`**
8. Directory - **`/srv/als-incoming`**
9. **Beginner** mode
10. **savedisk**
11. Image name - type a name with no spaces, e.g. `win11-office`
12. Source disk - pick the internal disk (`sda` or `nvme0n1`)
13. Accept the defaults for the remaining prompts, and confirm.

> **Capture to the staging folder, never straight to the library**
> `/srv/als-incoming` is writable; `/srv/als-images` is not. Capturing into staging is what stops a failed capture from replacing a good image that was working yesterday.

A 15 GB image takes roughly 15-25 minutes over gigabit. Leave it alone until it says it has finished.

Step 4 · Verify the capture before you trust it

This step exists because captures fail quietly. A full disk or an overheating server can end a capture part-way and still leave a folder that looks perfectly normal. Five minutes here saves finding out on a customer's machine.

Log in to the server and set the name you used:

```
ssh stephen01@192.168.0.20
IMG=win11-office
cd /srv/als-incoming/$IMG
```

a. Did it finish?

```
ls -l clonezilla-img disk parts *-pt.sf
tail -5 clonezilla-img
```

All four must exist. `clonezilla-img` is written last on a successful save, so its presence is the single best sign the capture ran to completion.

b. Any errors hidden in the log?

```
grep -inE 'no space|write error|not saved correctly|failed' clonezilla-img; echo
"exit=$?"
```

You want no lines printed and `exit=1`. Anything printed means recapture.

c. Is every partition actually there?

```
for p in $(cat parts); do ls -1 ${p}.*-img* 2>/dev/null || echo "MISSING: $p"; done
```

d. Is the data intact?

The definitive test. It decompresses every payload and checks the internal checksums, writing nothing. Several minutes on a 15 GB image.

```
zstd -t -T2 --no-progress *.zst; echo "exit=$?"
```

> **Keep the -T2**
> It limits the test to two CPU cores. Letting it use every core has previously pushed this server to 94 °C and triggered a thermal shutdown - which then looks exactly like a corrupt image. `exit=0` means intact. `premature end` means truncated; `Decoding error` means corrupt. Either way, recapture.

Step 5 · Publish it to the library

Only once every check above has passed:

```
sudo mv /srv/als-incoming/$IMG /srv/als-images/$IMG
sudo chown -R nobody:nogroup /srv/als-images/$IMG
ls -l /srv/als-images/
```

That is all that is required. The stations pick it up on their own - there is no manifest to edit. An image folder that is present and complete is offered under its folder name; one that is incomplete is not offered at all.

Step 6 · Give it a proper name

On any audit station, in **Load OS image**:

1. Pick the new image in **Select OS image**. It appears under its folder name, marked *not named yet*.
2. Click **Rename**.
3. Type the name that says what the image is - *Windows 11 Pro - Standard Office*.
4. Click **Save**.

That name is what the picker shows from then on. Name images by what they are **for**, not by their Windows version - three images all reading *Windows 11 Pro 23H2* tell you nothing about which is which.

> **Names are stored per stick**
> The name lives on the USB stick, because the image library is read-only. If you build a second audit stick, name the images on that one too. Clearing the field restores the original folder name.

Step 7 · Test it on one machine

Never put a brand new image into production without restoring it once.

1. Take a scrap machine of the type you will be imaging.
2. Boot the audit stick, select the drive and the new image, and click **Load OS Image**.
3. Wait. The panel shows GB written as it goes; a large NTFS partition can be quiet for a while at the start.
4. When it shows **Completed**, reboot the machine *without* the stick.
5. Confirm Windows reaches the out-of-box setup screen, the network works, and the display driver is correct.

> **What a good restore looks like**
> The status panel ends on **Completed** with the image name and target drive. If it ends on **Failed**, open **Show details** - the last output from Clonezilla, the decompressor and partclone is all captured there.

When something goes wrong

The restore sits at 0% and says nothing

Usually normal. Clonezilla can be silent for the whole of a large partition. The status line reports GB written straight from the kernel, so if that number is climbing it is working, whatever the percentage says. If you need certainty, press **Ctrl-Alt-F2** on the station and run:

```
while :; do awk '{printf "%s written: %.1f GB\n", strftime("%H:%M:%S"), $7/2097152}' \
/sys/block/nvme0n1/stat; sleep 5; done
```

Climbing means healthy. Flat for a minute or more means genuinely stuck. **Ctrl-C** to stop, **Ctrl-Alt-F1** to return.

"No space left on device" during capture

The server ran out of room. Delete old captures from `/srv/als-incoming` and start again. Check first with `df -h /srv`.

The capture stopped part-way for no obvious reason

Check whether the server overheated - this has happened before:

```
journalctl -k --no-pager | grep -iE 'critical temperature|thermal|mce' | tail -5
sensors
```

The image does not appear on the station

- It is still in `/srv/als-incoming` - move it to `/srv/als-images`.
- It is incomplete. An image without `disk`, `parts` and `clonezilla-img` is deliberately not offered.
- The station cannot reach the server. Check **Settings** → **Image server** reads `192.168.0.20:/srv/als-images`.

Restored Windows will not boot

Almost always a partition-only restore rather than a whole-disk one. The station only ever restores whole disks, so if you see this, the image itself was captured from a partition rather than a disk. Recapture using **savedisk**, not **saveparts**.

Quick reference

Everything above, condensed. `IMG` is your image folder name.

```
# on the server
ssh stephen01@192.168.0.20
IMG=win11-office

# 1. verify a fresh capture
cd /srv/als-incoming/$IMG
ls -l clonezilla-img disk parts *-pt.sf
grep -inE 'no space|write error|not saved correctly|failed' clonezilla-img
zstd -t -T2 --no-progress *.zst; echo "exit=$?"

# 2. publish it
sudo mv /srv/als-incoming/$IMG /srv/als-images/$IMG
sudo chown -R nobody:nogroup /srv/als-images/$IMG

# 3. check free space before the next one
df -h /srv
```

On the reference machine, before capturing:

```
C:\Windows\System32\Sysprep\sysprep.exe /generalize /oobe /shutdown
```

Clonezilla menu path: **Start_Clonezilla** → **device-image** → **nfs_server** → **dhcp** → **nfs** → **192.168.0.20** → **/srv/als-incoming** → **Beginner** → **savedisk**